

Review of Artificial Intelligence Techniques in Chronic Obstructive Lung Disease

Konstantinos P. Exarchos , Agapi Aggelopoulou, Aikaterini Oikonomou, Theodora Biniskou, Vasiliki Beli, Eirini Antoniadou, and Konstantinos Kostikas

Abstract—**Background:** Artificial Intelligence (AI) has proven to be an invaluable asset in the healthcare domain, where massive amounts of data are produced. Chronic Obstructive Pulmonary Disease (COPD) is a heterogeneous chronic condition with multiscale manifestations and complex interactions that represents an ideal target for AI. **Objective:** The aim of this review article is to appraise the adoption of AI in COPD research, and more specifically its applications to date along with reported results, potential challenges and future prospects. **Methods:** We performed a review of the literature from PubMed and DBLP and assembled studies published up to 2020, yielding 156 articles relevant to the scope of this review. **Results:** The resulting articles were assessed and organized into four basic contextual categories, namely: i) ‘COPD diagnosis’, ii) ‘COPD prognosis’, iii) ‘Patient classification’, iv) ‘COPD management’, and subsequently presented in an orderly manner based on a set of qualitative and quantitative criteria. **Conclusions:** We observed considerable acceleration of research activity utilizing AI techniques in COPD research, especially in the last couple of years, nevertheless, the massive production of large and complex data in COPD calls for broader adoption of AI and more advanced techniques.

Index Terms—Artificial intelligence, machine learning, data mining, chronic obstructive pulmonary disease, chronic bronchitis, emphysema.

I. INTRODUCTION

ARTIFICIAL Intelligence (AI) is a field of Computer Science that emerged around 1950 but has evolved substantially during the last decade with widespread adoption. AI mimics the human mind by developing machines able to process, learn and respond to information gained from data. It is specifically suited for tasks that involve large amounts of complex data, where the underlying patterns are elusive to the human mind. AI and its branches such as Machine Learning (ML) and Data Mining (DM) were early adopted in healthcare, but have been growing fast since the advent of genomic data and more recently

Manuscript received July 11, 2021; revised November 7, 2021; accepted December 12, 2021. Date of publication December 16, 2021; date of current version May 5, 2022. (Corresponding author: Konstantinos P. Exarchos.)

The authors are with the Respiratory Medicine Department, School of Medicine, University of Ioannina, 45500 Ioannina, Greece (e-mail: kexarcho@gmail.com; xristinaagg3@gmail.com; aikokonomou@hotmail.com; t.biniskou@gmail.com; vasilikibeli@hotmail.com; eirinantoniadou@gmail.com; kexarcho@gmail.com).

Digital Object Identifier 10.1109/JBHI.2021.3135838

with the broad availability of wearable sensors. Deep Learning (DL) is a subfield of Machine Learning that has been in the spotlight during the past few years and scales to vast amounts of complex data. In a recent review article regarding the adoption of AI in healthcare [1], it is noted that literature is primarily focused on cancer, neurological and cardiovascular diseases. AI research in respiratory diseases remains limited [2], even though the types of data used in respiratory medicine are suitable and the results gained so far from the available applications are quite encouraging [3]. The European Lung White Book [4] emphasizes on the importance and contribution of imaging data as well as genomic, metabolomic and proteomic data towards advancing respiratory research.

COPD is specifically distinguished among other respiratory diseases because it represents a major public health challenge and is a significant cause of chronic morbidity and mortality globally [5]. Throughout the course of COPD, a patient undergoes a multitude of examinations, e.g., spirometry, imaging studies, biomarker measurements, genetic tests, etc producing large amounts of data that capture the state and progression of the disease. Even though some of this information is well characterized and understood, there are several aspects of COPD that remain obscure and unexplained. AI can analyze the aforementioned data more efficiently and subsequently recognize underlying patterns in order to assist clinical decision making. Moreover, the increasing production of genomic data pertaining to COPD, coupled with sizeable data from imaging modalities that are produced in several time slices during the patient’s follow-up, makes AI analysis ideal for COPD related data. In this manuscript, we performed a review of the literature in order to summarize the applications of AI in Chronic Obstructive Pulmonary Disease (COPD).

Regarding AI techniques, starting from the bottom up, DM refers to the lower level coding steps for examining large data structures for meaningful patterns or trends. Building upon the patterns identified from DM, ML is the process of making a machine learn, i.e., increasing its performance in a certain task by assimilating more experience through data. DL is essentially a powerful class of ML, featuring a complex architecture based on Artificial Neural Networks of multiple non-linear layers. On top lies AI that emphasizes on the ability of machines to perform complex human tasks, e.g., observing, discerning and planning.

Roughly, every ML model undergoes two sessions or phases, namely the training phase where the model is trained to learn the

dependencies among the input variables; and the testing phase where new instances are presented to the aforementioned model in order to assess its performance and generalization capability. The learning procedure pertaining to the training phase can be performed either supervised or unsupervised. In supervised learning the algorithm is presented with a set of annotated input samples, i.e., samples where the class or target outcome is known. Whereas, in unsupervised learning the input samples are not annotated and the algorithm is expected to identify groupings within the input samples. There is also a hybrid type of learning called semi-supervised learning where the algorithm is fed with both annotated and unannotated samples.

After the training phase has concluded, the testing or evaluation phase aims to assess the performance of the trained model. For this purpose a subset of the initial dataset or a new independent dataset is presented to the trained model and the agreement between the reported and the actual class of the input samples is quantified. There are several partitionings that can be performed to the dataset in order to define the training and testing sets. Either an arbitrary percentage of samples is kept for testing (usually 10-40%, depending on the size of the dataset), or the dataset is divided in n equal subsets, whereby the $n-1$ are used for training and n th is used for testing and this is repeated n times; the latter process is called n -fold cross validation. Leave One Out Cross Validation (LOOCV) is a variation of n -fold cross validation where n is equal to the total number of samples. In order to quantify the performance of the ML model, several metrics can be calculated depending on the type of the employed ML algorithm. Accuracy and area under the receiver operating characteristics (ROC) curve (AUC) are two metrics that are frequently used in ML tasks; accuracy refers to the overall correctness of the model, i.e., the ratio of correctly predicted outcomes and the total number of samples, and AUC denotes the performance of the model at separating classes. In a methodologically similar review article discussing the adoption of AI techniques in asthma research [6], we provide in the supplement a more extensive introduction in AI techniques.

In the current review article, we searched the literature for articles utilizing AI/ML techniques in COPD research. In the 'Literature Review' section we present the steps followed in order to obtain all relevant research. Next, in the 'Artificial intelligence and COPD' section, our main findings from the literature review are presented, conceptually organized into four major categories, in order to facilitate reading and appraisal of the existing evidence.

II. LITERATURE REVIEW

In order to assemble the literature relevant to this work, two independent reviewers searched PubMed and DBLP, from inception until the end of December of 2020, for articles employing AI/ML techniques in COPD research. Specifically, we searched DBLP, which is a popular repository of Computer Science bibliography, using the keywords "COPD", "Emphysema" and "Chronic bronchitis". Next, we searched Pubmed using the following list of queries: ["Artificial Intelligence" AND COPD], ["Machine Learning" AND COPD], ["Data mining"

AND COPD], ["Deep Learning" AND COPD], ["Convolutional Neural Networks" AND COPD], ["Decision Tree" AND COPD], ["Neural Network" AND COPD], ["Support Vector Machine" AND COPD], ["Random Forest" AND COPD]; the latter query term in PubMed, i.e., 'COPD' was in turn replaced with the terms "Emphysema" and "Chronic bronchitis". The articles from the two repositories were merged and duplicates were removed. We maintained only journal articles presenting original research, written in English; the remaining articles were subsequently reviewed in order to maintain only relevant studies. Eventually, we maintained 156 articles that were grouped based on their content and purpose into four broad categories: (1) COPD diagnosis, (2) COPD prognosis, (3) Patient classification and (4) COPD management. The flowchart of these steps is presented in Fig. 1.

In Fig. 2 we present the distribution of articles conducting COPD research using AI/ML techniques from 2008 up to 2020. We observe a gradual increase in the adoption of AI/ML techniques over the course of approximately the past 10 years, with an identifiable peak in 2020.

III. ARTIFICIAL INTELLIGENCE AND COPD

In this section, we present the articles meeting the inclusion criteria for the current review article. In order to present our findings in a structured manner, we have defined four distinct categories and each article has been assigned in at least one category based on its content and purpose. Specifically, we have defined the following categories:

- 1) COPD diagnosis: in this category we have assigned articles that employ AI/ML techniques for diagnosing COPD.
- 2) COPD prognosis: this category contains articles assessing the prognosis of COPD patients, specifically focusing on the prediction of exacerbations or their prognosis.
- 3) Patient classification: studies that aim to classify or stratify COPD patients based on certain characteristics belong to this category.
- 4) COPD management: here we have assigned articles that utilize AI/ML techniques for facilitating the overall management of COPD.

In order to further facilitate reading and comprehension of the manuscript, in Table I we provide a list of some common abbreviations used in the AI/ML field that also appear in the rest of the manuscript. The first column contains some popular classification algorithms, and the second column contains techniques used for method validation or evaluation metrics used for quantifying algorithm performance. Moreover, we urge the reader to refer to a similar review [6] presented by our group where the supplement contains a primer on AI/ML techniques.

For each of the four categories we have created a summarizing table where the respective articles can be compared based on a series of qualitative and quantitative criteria. Specifically, those criteria are: AI/ML algorithm, sample size, input features, evaluation method and performance, and constitute the columns of each table. Due to their length, those tables are not included in

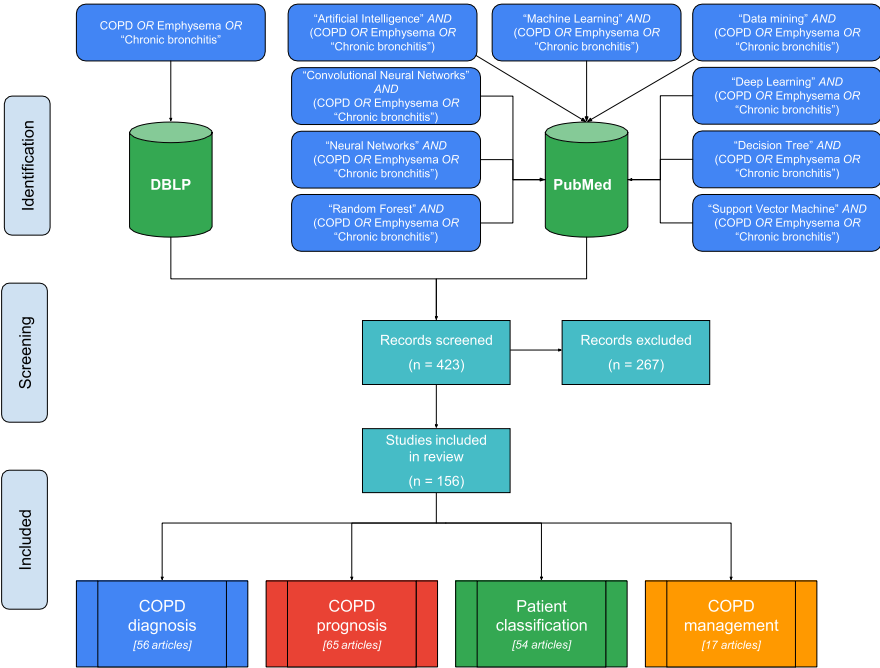


Fig. 1. Flowchart of the literature review.

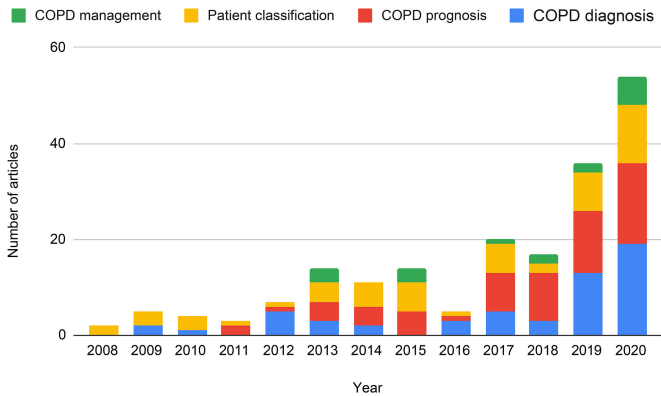


Fig. 2. Distribution of articles utilizing AI/ML techniques in COPD re-search.

the manuscript and are available from the authors upon request. In the article, we provide for each category an overview of the contained articles, as well as a brief description of the most recent and representative studies.

A. COPD Diagnosis

This category contains articles relevant to COPD diagnosis. There is no clear tendency for a specific machine learning algorithm; several popular algorithms are used for COPD diagnosis, such as Random Forests, Decision Trees, Convolutional Neural Networks, etc. As expected for the purpose of diagnosis, primarily clinical data have been employed in the majority of the studies. Symptoms, medical history, pulmonary function tests,

TABLE I
LIST OF COMMON AI/ML ABBREVIATIONS

Classification algorithms		Evaluation techniques	
<i>BN</i>	Bayes Network	<i>CV</i>	Cross Validation
<i>NB</i>	Naive Bayes	<i>LOOCV</i>	Leave One Out Cross Validation
<i>SVM</i>	Support Vector Machine	<i>Se</i>	Sensitivity
<i>k-NN</i>	k Nearest Neighbors	<i>Sp</i>	Specificity
<i>DT</i>	Decision Tree	<i>Acc</i>	Accuracy
<i>RF</i>	Random Forest	<i>FPR</i>	False Positive Rate
<i>LR</i>	Logistic Regression	<i>ROC</i>	Receiver Operating Characteristic
<i>CNN</i>	Convolutional Neural Network	<i>AUC</i>	Area Under ROC Curve

physical examination findings are some of the features comprising this type of input (i.e., clinical). Though, it should be noted that imaging data are being increasingly used especially during the last couple of years, in conjunction with Deep Learning techniques. Regarding sample size, we observe great variations in the number of subjects or samples used in each study, ranging from approximately 30 up to several hundred thousand. In terms of evaluation, variations of n-fold cross validations have been used in the majority of the studies. Interestingly, out of the 56 studies in this category, approximately 20 were published in 2020, half of which employed Deep Learning algorithms. We

have selected a few of the most recent and representative studies that are briefly presented hereafter.

Tang *et al.* [7] analyzed large sets of inspiratory CT scans from ex-smokers and current smokers for the automated detection of COPD. They proposed a software pipeline based on Deep Learning that was validated in an external dataset yielding an AUC of 0.886. It is noteworthy that training was performed on subjects recruited as part of a lung cancer screening protocol, indifferent to their lung function testing, which might involve bias.

Mostafei *et al.* [8] explored the expression of genes in airway epithelial cells using several ML algorithms, aiming to identify a gene subset that is associated with COPD. The authors studied a population of 133 patients (healthy non-smokers, healthy smokers and COPD smokers) and came down to 44 genes resulting in $AUC = 0.82$; these genes could also serve as novel targets for therapeutic interventions. Studies employing genomic data (e.g. gene expression) could benefit significantly from AI/ML algorithms that are specifically suitable for analyzing large amounts of complex data with composite interactions among them.

Leidy *et al.* [9] analyzed three large datasets using Random Forests, in order to identify the key clinical features that differentiate certain patient subsets. Specifically four scenarios were examined: 1) moderate to severe COPD versus no COPD, 2) undiagnosed versus diagnosed COPD, 3) COPD with and without exacerbation history, and 4) clinically significant COPD versus all others. Across all scenarios the following features were used in the best models: age, smoking status/history, symptoms, activity limitation, episodes of acute bronchitis, and/or missed work days and non-work activities.

In early 2019 Topalovic *et al.* [3] published an article that attracted considerable attention; in their extensively validated work they proposed an AI-based algorithm with a twofold purpose: i) the interpretation of pulmonary function tests (PFT) and ii) the assignment of correct diagnosis out of a predefined set. The proposed algorithm identified the PFT perfectly and assigned correct diagnosis in 82% of the cases, whereas the pulmonologists yielded inferior performance in both tasks. This article shows an interesting and successful implementation of AI algorithms in everyday practice of respiratory physicians.

Cheng *et al.* [10] mined large scale medical records aiming to identify risk patterns in consecutive patient visits. The extracted risk patterns are subsequently fed to a classification algorithm in order to early diagnose a certain disease. They demonstrated the effectiveness of their method for the early assessment of COPD. Such methods can be incorporated in an electronic medical record and provide meaningful insight by running in the background. The model utilizes several time slices across the patients' follow up visits and captures temporal information which achieves enhanced information yet with increased complexity.

Spathis and Vlamos [11] implemented an AI-based Decision Support System that exploits several clinical features (e.g., age, sex, sputum production, smoking, etc) as input aiming to diagnose asthma and COPD. For the case of COPD the best performance was achieved using a Random Forest classifier

that yielded Precision of 98%, using a set of trivial and easily obtainable characteristics.

B. COPD Prognosis

This category contains articles that employ machine learning algorithms in order to assess the prognosis of patients with COPD; this primarily involves the early prediction of exacerbations as well as several characteristics pertaining to them, such as severity, hospital admission, mortality, etc. Same as in the previous category, clinical data are employed in the vast majority of the studies, with only a few studies using genomic, environmental or imaging data. It should be noted that a couple of studies have employed metabolomic data, which seems like an interesting prospect for future studies in the field of COPD research. In this category, Artificial Neural Networks have been utilized in several studies, and other popular machine learning algorithms such as Decision Trees or Random Forests are used to a lesser extent. In terms of evaluation and performance, n-fold cross validation has been used in the great majority of the studies from this category, and it is interesting that almost all works report quite encouraging results, with high AUC and accuracy. Some of the most notable and recently published studies from this group are described in the paragraphs that follow.

Ying *et al.* [12] proposed an automatic classifier, based on Deep Learning, for exacerbation risk assessment in patients diagnosed with COPD. The authors used patients from the COPDGene cohort, containing more than 10,000 patients, and after a multi-step feature selection process retaining the most informative features, they trained a Deep Belief Network achieving accuracy of 91.99%.

In their paper, Swaminathan *et al.* [13] have analyzed using several machine learning algorithms a wide range of variables regarding the patient profile, comorbidities, symptoms and vital signs of COPD patients, with a twofold purpose: i) to identify acute exacerbations of COPD and ii) performing triage regarding the severity of a potential exacerbation. A large set containing approximately 2500 cases was employed, where 101 cases were used for validation. After extensive experiments the authors came down to two models, one for triage and the other for predicting exacerbations, achieving performance superior compared to a set of board trained physicians, in both cases yielding overall accuracy above 90%.

In a recently published article [14] Goto *et al.* analyzed approximately 45,000 hospitalizations for COPD exploiting administrative data from a nationwide database in Japan. They used a set of variables pertaining to the hospital stay and the patient profile, aiming to predict hospital readmission within 30 days after discharge. They compared the reference model (logistic regression) with two machine learning algorithms (lasso regression and deep neural network) which achieved higher discrimination ability. Tube feeding duration, blood transfusion, thoracentesis use, and male sex were reported as important predictors of hospital readmission.

The next article from Merone *et al.* [15] focuses on tele-monitoring systems that have received considerable criticism especially when applied to health oriented applications, due to the

lack of in person evaluation of the patient. Nevertheless, it should be highlighted that COPD patients, especially in the last stages where long-term oxygen therapy is needed, are unable to visit the respiratory physician regularly, therefore a tele-monitoring system might prove quite beneficial particularly in COPD. In this work, a tele-monitoring system is applied that collects a series of easily attainable physiological data (i.e., oxygen saturation and heart rate) and subsequently the authors build a Decision Support System that prompts for worrisome events, e.g., exacerbation. The proposed system achieves overall accuracy 98.4% which is better than the one achieved by the medical experts. It should be noted however, that the study included 22 patients, therefore further validation in richer datasets is needed, in order to utilize such a system in clinical practice.

Shah *et al.* [16] proposed a telemonitoring system based on a digital health platform recording the patient status via several self-monitoring modules (e.g., questionnaires), as well as a set of vital signs acquired from a pulse oximeter. Their aim was to develop a robust telemonitoring system for tracking the progression of the disease and primarily exploit this information in order to early identify COPD exacerbations. The proposed algorithm was validated using approximately 27,000 sessions from 110 patients, yielding satisfactory results in terms of sensitivity, though lacking in specificity.

The article by Gonzalez *et al.* [17] employs Deep Learning which has proven to be an important asset in the healthcare domain, especially for the analysis of large amounts of complex imaging modalities. The authors utilize a CNN, which is probably the most popular deep learning-based model, and analyze Computed Tomography imaging data aiming to diagnose COPD and moreover capture its prognosis. The proposed algorithm yielded a C-statistic of 0.86 for the detection of COPD, whereas the respective C-statistic for predicting acute respiratory disease was between 0.55 and 0.64.

C. Patient Classification

The studies comprising this category aim to stratify patients into certain categories, based on a set of characteristics. Some studies, primarily the ones employing clinical data stratify patients in terms of disease severity, prognosis or even cost. It should be noted, however, that a significant portion of the studies in this category focus on imaging modalities, predominantly CT scans, aiming to assess characteristics that differentiate patients, e.g., emphysema. To this end, several studies utilizing Deep Learning algorithms have been published especially in 2019 and 2020. Regarding the sample size, we observe that overall, compared to the other categories, fewer subjects are used and consequently, the leave one out cross validation technique is used, which is specifically suited for small datasets. Same as before, we have selected a few studies from this category that we briefly describe.

Peng *et al.* [18] present a multi-scale residual network with two channels of raw CT image and its differential excitation component, aiming to classify and quantify emphysema, yielding an overall accuracy of 93.74%. Moreover, the authors assessed the correlation between emphysema and pulmonary

functions, conjecturing that centrilobular and panlobular emphysema have strong correlation with pulmonary function indices.

Moghadas-Dastjerdi *et al.* [19] propose an automatic COPD assessment technique based on the analysis of imaging modalities. The authors extracted 23 features of air volume variation and distribution from CT scans of 69 patients with COPD and aimed to correlate their GOLD disease stage with findings on CT scans. Using a subset of 12 features coupled with the Naive Bayes classifier, the proposed models achieved an overall accuracy of 84%.

Due to the large heterogeneity inherent to COPD in its aetiology, phenotypes and prognosis, Pikoula *et al.* [20] proposed an unsupervised classification scheme in order to identify clinically meaningful patient subgroups. They used as input 16 clinical features (e.g., risk factors, comorbidities) and analyzed a set of approximately 30,000 subjects acquired from the national electronic health record of England. Five COPD patient clusters were identified with the following leading characteristics: 1) anxiety/depression; 2) severe airflow obstruction and frailty; 3) cardiovascular disease and diabetes, 4) obesity/atopy, and 5) low prevalence of comorbid conditions.

Lin *et al.* [21] developed a smooth Bayesian Network to identify future high cost patients with COPD, thus, tailoring appropriate individualized interventions that prevent exacerbations and decelerate disease progression. The study used the medical insurance data of 186,188 patients diagnosed with COPD over a 3-year period, and achieved AUC of 0.80 towards predicting high cost patients. Notably, the authors identified region to be a significant factor pertaining to the cost of a specific patient.

Pinto-Plata and colleagues [22] extracted a series of plasma metabolites that were subsequently fed to an SVM aiming to stratify patients diagnosed with severe COPD into survivors and non-survivors based on a 2-year follow-up. The proposed methodology was validated in a set of 90 COPD patients and 30 controls, yielding overall accuracy of 85%. The authors report a plasma metabolomic profile characterized by an oxidative energy production difference between survivors and non-survivors, that can be detected in COPD patients 2 years before death.

D. COPD Management

This last category, which is the least populated one, contains studies that focus on the treatment of patients with COPD, as well as the formulations of their management plan. Almost all studies rely on clinical data and employ a variable number of subjects, ranging from only 10 up to hundreds of thousands; it should be noted that such huge datasets often contain administrative data from hospitals or insurance companies, with all the potential caveats that these datasets have. Same as in the previous categories, n-fold cross validation has been employed for evaluation purposes in the majority of the studies. There is only one study utilizing a Deep Learning algorithm that was published in 2020; it should also be noted that studies published in 2020 constitute a considerable proportion of the studies in this category, denoting possibly an increasing interest. Just below, we briefly present a couple of works pertaining to the category of ‘COPD management’.

Same as several other studies reported previously, the work by Lee *et al.* [23], aims to identify COPD readmission using a Bayesian Network model. Subsequently, the authors describe the utilization of such a model for designing and fine-tuning intervention plans to avoid readmissions. In a similar manner, studies that aim to early predict acute exacerbations of COPD [15], [16] or identify high risk/cost [21], [22] patients also fall within this category as they can be exploited for formulating a personalized COPD management plan.

Weng *et al.* [24] extracted 8 features, namely: gender, age, illness duration, respiratory rate, arterial blood gases and plasma albumin content, that were subsequently fed as input to a machine learning algorithm. Their purpose was to identify the patients that were more likely to benefit from non-invasive positive pressure ventilation and also pinpoint the factors that dictate this decision. The proposed model was evaluated on three different datasets, yielding an overall accuracy around 90%.

In the literature the therapy and overall management of chronic bronchitis remains under debate. In [25] the authors proposed a modification of the fuzzy C-means algorithm by exploiting the “kernel trick” whereby the initial samples are mapped to a higher dimensionality feature space. The algorithm was evaluated on a set of patients with chronic bronchitis aiming to optimize their treatment; the proposed kernel fuzzy C-means algorithm resulted in a 10% increase in overall accuracy compared to the standard fuzzy C-means algorithm.

IV. DISCUSSION

The most notable observation is the increased adoption of AI techniques in COPD research; the same trend was also observed in a recent review conducted by our group in asthma research [6], both in accordance with the overall trend in the healthcare domain [1]. It should be noted that AI applications in asthma and COPD exhibit only little overlap, even though they both refer to obstructive lung diseases; one key difference lies in the utilization and analysis of imaging data. COPD patients can be accompanied by a lot of imaging data especially CT scans, whereas asthma management rarely depends on imaging data. Moreover, COPD diagnosis is pretty straightforward compared to asthma that does not conform to strict criteria. Overall, respiratory research is gradually picking up on AI techniques with a slight delay compared to other healthcare domains such as cancer or cardiovascular research [26], [27].

We also observed an odd gap in articles employing AI for COPD research during 2016; from 2008 and up to 2020 there is a gradual increase in such studies with a sudden drop of interest in 2016. Since the number of studies over those years is around 10 per year this drop could merely be attributed to the small sample size. What is more important, is the remarkable increase of relevant studies in 2019 and 2020 by a factor of 3 and 4 respectively, compared to the previous years.

There are certain aspects pertaining to COPD research that can benefit greatly from the utilization of AI techniques. First of all, COPD patients might undergo several CT scans primarily of the thorax, over the course of their follow-up, for various reasons, e.g., nodules, emphysema, bronchiectasis, pulmonary

embolism, fibrosis, etc [28]. However, it should be noted that CT scans are not routinely recommended in the clinical assessment of COPD, but may be helpful in the differential diagnosis where concomitant diseases are present. Such large and complex imaging modalities can be effectively analyzed using AI techniques, yielding results far superior to human users [29], [30]. Moreover, some of the aforementioned tasks need to identify subtle changes over consecutive modalities in a person to person basis, which is particularly difficult for the human eye.

The utilization of AI techniques in the analysis of imaging data has attracted particular attention and is rapidly evolving [31], [32]. CT scans play an integral role in respiratory medicine overall, and in the current literature review, there is a large body of studies in this field. Specifically, among the 156 reviewed articles, 52 employ imaging data in their analysis, the vast majority of which CT scans. Most of these studies aim to detect and quantify emphysema, and often achieve satisfactory performance, especially the latest ones [12], [33]. Some of these studies employ large patient cohorts, e.g., from COPDGene, thus, achieving fair generalization. There are also a couple of works that subsequently aim to correlate the type and extent of emphysema with pulmonary functions tests [18], [34], probably in an attempt to fuse CT scans with established indices of COPD management. It is also worth noting that Deep Learning based algorithms for the analysis of imaging data are increasingly used from 2017 and on, constituting the majority of the studies presented in 2020; whereas, in the earlier studies SVMs were employed more frequently.

Besides imaging data, the evolution of variable COPD characteristics can be detected from the molecular level, where tremendous heterogeneity is attributed to genetic interactions, to the patient level as a whole as well as to the interaction of the patient with the environment. All these changes take place over a person’s lifetime depicting a personalized timeline of the disease where multiscale and multiparametric time-course data capture the natural history and progression of the disease. These complex interactions ranging in time and dimension exceed the capabilities of mere statistical models and are ideal for advanced machine learning algorithms. However, it should be noted that since these algorithms are primarily data-driven, training them effectively requires enormous amounts of well characterized data.

The surge of data in massive quantities and great complexity produced in healthcare and COPD research in particular, oftentimes exceeds the human limits, and calls for increased reliance on artificial intelligence, and the Deep Learning subtype in particular. Deep learning involves training an Artificial Neural Network with many layers, using as input vast amounts of raw or minimally preprocessed data. Though they require less human guidance, and often achieve remarkable results, Deep Learning algorithms require enormous amounts of data to capture the underlying complexity. Deep learning algorithms have brought about promising results in complex diagnostics such as ophthalmology [35], pathology [36], dermatology [37], etc. In the current review article, besides imaging data, there are a few more interesting applications of Deep Learning in COPD research. Altan *et al.* [38] analyzed multi-channel lung sounds for the early diagnosis of COPD, achieving overall accuracy of 93.67%.

In a similar manner, García-Ordás *et al.* [39] proposed a CNN trained from lung sounds for the identification of various respiratory pathologies. Another interesting approach is presented by Tang *et al.* [40] who generated a temporal visualization of COPD progression based on unstructured clinical notes.

Moreover, it has been established that COPD does not have a unanimous presentation and evolution in all patients [41], therefore, cluster-based or even personalized approaches are likely to perform better. Similarly, such approaches need massive amounts of data in order to capture individualized changes, and also call for algorithms that are able to perform online training, i.e., a method of machine learning where that algorithm can dynamically adapt to new instances and improve over time [42]. This is particularly important in the development of algorithms that may predict the occurrence and timing of future events, especially those requiring healthcare resource utilization (e.g., exacerbations and hospitalizations). Such approaches may support early intervention to the benefit of patients.

Another field of COPD and respiratory research in general that has attracted considerable attention lately is spirometry and pulmonary function tests. As described earlier in the manuscript, Topalovic *et al.* [3] analyzed pulmonary function tests with machine learning algorithms achieving perfect discrimination of the pre-specified categories, thus, outperforming a group of pulmonologists in the same task. Further applications of AI may support the use of spirometric results combined with clinical characteristics to support diagnostic processes, including the differential diagnosis between asthma and COPD. Such applications of AI when they become available may provide useful aids to physicians supporting their daily clinical practice.

The necessity of AI in COPD research and its subsequent spread have been discussed elsewhere in this article. There are a few more features of AI algorithms that make them appealing for healthcare applications; they are less fallible to fatigue, they can produce rapid decisions and are not affected by the urgent pressure of the clinical setting. Moreover, AI algorithms can process and assimilate large amounts of patients, exceeding the comprehension of a physician in an entire lifetime. The diversity of these input patient sets can lead to decreased bias that can also be found in human physicians. Despite all the promises of AI technology, there are also several limitations, pitfalls and considerations that need to be taken into account, especially in the sensitive field of healthcare. The ‘black-box’ nature of AI algorithms has long been an issue of controversy, which has become even more relevant with Deep Learning. Nevertheless, technology and AI algorithms in particular, aim to support the work of physicians rather than substitute them, therefore, the dilemma for physicians is whether or not to accept this extra helping hand. One might argue that this opaqueness already exists in medical practice where prescription drugs are administered without knowing the mechanism of action. One of the major issues regarding AI algorithms acceptance is that normally clinical knowledge is not embedded in the algorithms, but they rather rely on the quality of data, conforming to the cliché “garbage in, garbage out”. Another important issue is that currently the predictions of AI algorithms are fragmented in nature, e.g., an algorithm detecting pneumonia in a chest X-ray

might achieve performance comparable (or even superior) to a human, yet a radiologist can conjecture a lot more information from a chest X-ray besides an infiltrate. Arguably, there is a lot more in the patient-physician interaction, such as empathy, trust and instinct, that can not be replaced.

Eventually, an emerging field in COPD research that is expected to grow significantly in the next few years is the personalized management of COPD patients, including the adoption of biologic treatments in the treatment of COPD. Due to their inherent complexity, their interaction in metabolic pathways under exploration, and the matching of each biologic treatment with the patient subset that is expected to have the best effectiveness, AI is expected to play an important role in this field as well.

The development of AI-based health applications and their implementation at the patient level, e.g., using wearables packed with sensors, is becoming more and more popular. Enabling the public to take healthcare in their own hands should be realized with caution, especially when delivered unattended by physicians. For example a smartwatch detecting atrial fibrillation is prone to false positives that might lead to unnecessary medical evaluations; this could further escalate when applied to large sets of low-risk, young patients. Specifically, for the case of COPD, sensors such as SpO₂, respiratory rate, etc are becoming more accurate and widespread. In a similar manner, no-code or low-code AI tools, especially for Deep-Learning are gaining increasing popularity in several fields including healthcare. Though this might lead to a surge of interesting applications, so will faulty and poorly performing algorithms, therefore, rigorous studies and a concrete validation process prior to approval is essential.

V. CONCLUSION

In the last couple of years AI is rapidly invading respiratory research, following its broad adoption in other healthcare domains. Besides imaging modalities often accompany COPD patients, modern medical devices (spirometers, oscillometers, ventilation devices, sensors, wearables, etc) and genomic information becoming more readily available, comprise a multiplex of data that calls for advanced AI algorithms. To this end, the most prominent challenge is to synthesize rich datasets that are rigorously annotated, in order to produce effective algorithms with generalization capability. These conditions are prerequisite for the development of reliable models that will assist with the delivery of better care to COPD patients.

REFERENCES

- [1] F. Jiang *et al.*, “Artificial intelligence in healthcare: Past, present and future,” *Stroke Vasc Neurol.*, vol. 2, no. 4, pp. 230–243, Dec. 2017.
- [2] E. Angelini, S. Dahan, and A. Shah, “Unravelling machine learning: Insights in respiratory medicine,” *Eur. Respir. J.*, vol. 54, no. 6, pp. 1901216, 2019. doi: [10.1183/13993003.01216-2019](https://doi.org/10.1183/13993003.01216-2019).
- [3] M. Topalovic *et al.*, “Artificial intelligence outperforms pulmonologists in the interpretation of pulmonary function tests,” *Eur. Respir. J.*, vol. 53, no. 4, Apr. 2019, doi: [10.1183/13993003.01660-2018](https://doi.org/10.1183/13993003.01660-2018).
- [4] G. John Gibson, R. Loddenkemper, Y. Sibille, and B. Lundbäck, *The European Lung White Book: Respiratory Health and Disease in Europe*. Sheffield, U.K., European Respiratory Society, 2013.

- [5] Global initiative for chronic obstructive lung disease, pocket guide to COPD diagnosis, management and prevention: A guide for healthcare. 2017.
- [6] K. P. Exarchos, M. Beltsiou, C.-A. Votti, and K. Kostikas, "Artificial intelligence techniques in asthma: A systematic review and critical appraisal of the existing literature," *Eur. Respir. J.*, May 2020, doi: [10.1183/13993003.00521-2020](https://doi.org/10.1183/13993003.00521-2020).
- [7] L. Y. W. Tang, H. O. Coxson, S. Lam, J. Leipsic, R. C. Tam, and D. D. Sin, "Towards large-scale case-finding: Training and validation of residual networks for detection of chronic obstructive pulmonary disease using low-dose CT," *Lancet Digit. Health*, vol. 2, no. 5, pp. e259–e267, May 2020.
- [8] S. Mostafaei *et al.*, "Identification of novel genes in human airway epithelial cells associated with chronic obstructive pulmonary disease (COPD) using machine-based learning algorithms," *Sci. Rep.*, vol. 8, no. 1, p. 15775, Oct. 2018.
- [9] N. K. Leidy *et al.*, "Insight into best variables for COPD case identification: A random forests analysis," *Int. J. Chron. Obstruct. Pulmon. Dis.*, vol. 3, no. 1, pp. 406–418, 2016.
- [10] Y.-T. Cheng, Y.-F. Lin, K.-H. Chiang, and V. S. Tseng, "Mining sequential risk patterns from large-scale clinical databases for early assessment of chronic diseases: A case study on chronic obstructive pulmonary disease," *IEEE J. Biomed. Health Informat.*, vol. 21, no. 2, pp. 303–311, Mar. 2017.
- [11] D. Spathis and P. Vlamos, "Diagnosing asthma and chronic obstructive pulmonary disease with machine learning," *Health Informat. J.*, vol. 25, no. 3, pp. 811–827, Sep. 2019.
- [12] J. Ying *et al.*, "Classification of exacerbation frequency in the COPDGene cohort using deep learning with deep belief networks," *IEEE J. Biomed. Health Informat.*, vol. 24, no. 6, pp. 1805–1813, Jun. 2020.
- [13] S. Swaminathan *et al.*, "A machine learning approach to triaging patients with chronic obstructive pulmonary disease," *PLoS One*, vol. 12, no. 11, Art. no. e0188532, Nov. 2017.
- [14] T. Goto, T. Jo, H. Matsui, K. Fushimi, H. Hayashi, and H. Yasunaga, "Machine learning-based prediction models for 30-day readmission after hospitalization for chronic obstructive pulmonary disease," *COPD*, vol. 16, no. 5/6, pp. 338–343, Dec. 2019.
- [15] M. Merone, C. Pedone, G. Capasso, R. A. Incalzi, and P. Soda, "A decision support system for tele-monitoring COPD-related worrisome events," *IEEE J. Biomed. Health Informat.*, vol. 21, no. 2, pp. 296–302, Mar. 2017.
- [16] S. A. Shah, C. Velardo, A. Farmer, and L. Tarassenko, "Exacerbations in chronic obstructive pulmonary disease: Identification and prediction using a digital health system," *J. Med. Internet Res.*, vol. 19, no. 3, pp. e69, Mar. 2017.
- [17] G. González *et al.*, "Disease staging and prognosis in smokers using deep learning in chest computed tomography," *Amer. J. Respir. Crit. Care Med.*, vol. 197, no. 2, pp. 193–203, Jan. 2018.
- [18] L. Peng *et al.*, "Classification and quantification of emphysema using a multi-scale residual network," *IEEE J. Biomed. Health Informat.*, vol. 23, no. 6, pp. 2526–2536, Nov. 2019.
- [19] H. Moghadas-Dastjerdi *et al.*, "Lung CT image based automatic technique for COPD GOLD stage assessment," *Expert Syst. with Appl.*, vol. 85, pp. 194–203, 2017, doi: [10.1016/j.eswa.2017.05.036](https://doi.org/10.1016/j.eswa.2017.05.036).
- [20] M. Pikoula, J. K. Quint, F. Nissen, H. Hemingway, L. Smeeth, and S. Denaxas, "Identifying clinically important COPD sub-types using data-driven approaches in primary care population based electronic health records," *BMC Med. Informat. Decis. Mak.*, vol. 19, no. 1, p. 86, Apr. 2019.
- [21] S. Lin, Q. Zhang, F. Chen, L. Luo, L. Chen, and W. Zhang, "Smooth bayesian network model for the prediction of future high-cost patients with COPD," *Int. J. Med. Informat.*, vol. 126, pp. 147–155, Jun. 2019.
- [22] V. Pinto-Plata *et al.*, "Plasma metabolomics and clinical predictors of survival differences in COPD patients," *Respir. Res.*, vol. 20, no. 1, p. 219, Oct. 2019.
- [23] S. Lee *et al.*, "Reducing COPD readmissions: A causal Bayesian network model," *IEEE Robot. Automat. Lett.*, vol. 3, no. 4, pp. 4046–4053, Jul. 2018, doi: [10.1109/Ira.2018.2861084](https://doi.org/10.1109/Ira.2018.2861084).
- [24] Y. Weng, Y. Fang, H. Yan, Y. Yang, and W. Hong, "Bayesian non-parametric classification with tree-based feature transformation for NIPPV efficacy prediction in COPD patients," *IEEE Access*, vol. 7, pp. 177774–177783, Dec. 2019, doi: [10.1109/access.2019.2958047](https://doi.org/10.1109/access.2019.2958047).
- [25] W. Lu and Z. Yan, "An improved fuzzy C-means clustering algorithm for assisted therapy of chronic bronchitis," *Technol. Health Care*, vol. 23, no. 6, pp. 699–713, 2015.
- [26] K. Kourou, T. P. Exarchos, K. P. Exarchos, M. V. Karamouzis, and D. I. Fotiadis, "Machine learning applications in cancer prognosis and prediction," *Comput. Struct. Biotechnol. J.*, vol. 13, pp. 8–17, 2015.
- [27] K. Kourou, K. P. Exarchos, C. Papaloukas, P. Sakaloglou, T. Exarchos, and D. I. Fotiadis, "Applied machine learning in cancer research: A systematic review for patient diagnosis, classification and prognosis," *Comput. Struct. Biotechnol. J.*, vol. 19, pp. 5546–5555, Oct. 2021.
- [28] W. W. Labaki *et al.*, "The role of chest computed tomography in the evaluation and management of the patient with chronic obstructive pulmonary disease," *Amer. J. Respir. Crit. Care Med.*, vol. 196, no. 11, pp. 1372–1379, Dec. 2017.
- [29] Y. Xu *et al.*, "Deep learning predicts lung cancer treatment response from serial medical imaging," *Clin. Cancer Res.*, vol. 25, no. 11, pp. 3266–3275, Jun. 2019.
- [30] M. Hardy and H. Harvey, "Artificial intelligence in diagnostic imaging: Impact on the radiography profession," *Br. J. Radiol.*, vol. 93, no. 1108, pp. 20190840, Apr. 2020.
- [31] O. Oren, B. J. Gersh, and D. L. Bhatt, "Artificial intelligence in medical imaging: Switching from radiographic pathological data to clinically meaningful endpoints," *Lancet Digit. Health*, vol. 2, no. 9, pp. e486–e488, Sep. 2020.
- [32] A. S. Panayides *et al.*, "AI in medical imaging informatics: Current challenges and future directions," *IEEE J. Biomed. Health Informat.*, vol. 24, no. 7, pp. 1837–1857, Jul. 2020.
- [33] C. Xu *et al.*, "DCT-MIL: Deep CNN transferred multiple instance learning for COPD identification using CT images," *Phys. Med. Biol.*, vol. 65, no. 14, p. 145011, Jul. 2020.
- [34] A. M. Fischer *et al.*, "Comparison of artificial intelligence-based fully automatic chest CT emphysema quantification to pulmonary function testing," *AJR Amer. J. Roentgenol.*, vol. 214, no. 5, pp. 1065–1071, May 2020.
- [35] J. De Fauw *et al.*, "Clinically applicable deep learning for diagnosis and referral in retinal disease," *Nat. Med.*, vol. 24, no. 9, pp. 1342–1350, Sep. 2018.
- [36] G. Litjens *et al.*, "Deep learning as a tool for increased accuracy and efficiency of histopathological diagnosis," *Sci. Rep.*, vol. 6, pp. 26286, May 2016.
- [37] A. Esteva *et al.*, "Dermatologist-level classification of skin cancer with deep neural networks," *Nature*, vol. 542, no. 7639, pp. 115–118, Feb. 2017.
- [38] G. Altan, Y. Kutlu, and N. Allahverdi, "Deep learning on computerized analysis of chronic obstructive pulmonary disease," *IEEE J. Biomed. Health Informat.*, vol. 24, no. 5, pp. 1344–50, Jul. 2019, doi: [10.1109/JBHI.2019.2931395](https://doi.org/10.1109/JBHI.2019.2931395).
- [39] M. T. García-Ordás *et al.*, "Detecting respiratory pathologies using convolutional neural networks and variational autoencoders for unbalancing data," *Sensors*, vol. 20, no. 4, Feb. 2020, doi: [10.3390/s20041214](https://doi.org/10.3390/s20041214).
- [40] C. Tang *et al.*, "A temporal visualization of chronic obstructive pulmonary disease progression using deep learning and unstructured clinical notes," *BMC Med. Informat. Decis. Mak.*, vol. 19, no. 8, p. 258, Dec. 2019.
- [41] G. Viegi, F. Pistelli, D. L. Sherrill, S. Maio, S. Baldacci, and L. Carrozzi, "Definition, epidemiology and natural history of COPD," *Eur. Respir. J.*, vol. 30, no. 5, pp. 993–1013, Nov. 2007.
- [42] P.-N. Tan, M. Steinbach, A. Karpatne, and V. Kumar, *Introduction to Data Mining*. Reading, MA: Addison-Wesley, 2013.